

IB Mathematics AA HL

Comprehensive Formula Sheet, Derivations, and Exam Guide

Built from the uploaded AA HL syllabus/guide structure and extended with exam-method notes

How to use this handbook. Learn the formula, then the derivation, then the trigger that tells you when to use it. In IB AA HL, marks are often lost not because the mathematics is hard, but because the method is incomplete, notation is sloppy, or the final line is not tied back to the question.

1 What the course actually demands

The AA HL course is built around five syllabus areas: Number and Algebra, Functions, Geometry and Trigonometry, Statistics and Probability, and Calculus. The official guide also sets out the assessment structure: Paper 1 without a calculator, Paper 2 with a calculator, Paper 3 extended response, and the Internal Assessment. The exam reward pattern is simple: correct setup, valid mathematical method, and clear communication matter as much as the final answer.

Full-mark habit. Write enough working that each transformation is justified. State domains when inverses or logarithms are involved. For equations, finish by checking extraneous roots and giving the solution set in the form the question asks for.

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2 Number and Algebra

2.1 Scientific notation and numerical manipulation

A number in scientific notation is written as $a \times 10^k$ where $1 \leq a < 10$ and $k \in \mathbb{Z}$. This is not cosmetic; it prevents scale errors in modelling and estimation.

$$(a \times 10^m)(b \times 10^n) = (ab) \times 10^{m+n}, \quad \frac{a \times 10^m}{b \times 10^n} = \frac{a}{b} \times 10^{m-n}$$

Exam note: keep the mantissa between 1 and 10. If your calculator gives 7.2×10^{-3} , do not rewrite it as 72×10^{-4} unless the question explicitly wants a different form.

2.2 Arithmetic sequences and series

An arithmetic sequence has constant difference d .

$$u_n = u_1 + (n-1)d, \quad S_n = \frac{n}{2}(2u_1 + (n-1)d) = \frac{n}{2}(u_1 + u_n)$$

Derivation of the sum. Write the sum forwards and backwards:

$$S_n = u_1 + u_2 + \cdots + u_n$$

$$S_n = u_n + u_{n-1} + \cdots + u_1$$

Add:

$$2S_n = (u_1 + u_n) + (u_2 + u_{n-1}) + \cdots$$

Each pair is $u_1 + u_n$, repeated n times, so $2S_n = n(u_1 + u_n)$.

Paper 1 trap. If you are asked to find a term number, isolate n algebraically before substituting numbers. Do not guess from pattern.

2.3 Geometric sequences and series

A geometric sequence has constant ratio r .

$$u_n = u_1 r^{n-1}, \quad S_n = \frac{u_1(r^n - 1)}{r - 1} = \frac{u_1(1 - r^n)}{1 - r}, \quad r \neq 1$$

$$S_\infty = \frac{u_1}{1 - r}, \quad |r| < 1$$

Derivation of the finite sum. Let $S_n = u_1 + u_1 r + \cdots + u_1 r^{n-1}$. Multiply by r :

$$rS_n = u_1 r + u_1 r^2 + \cdots + u_1 r^n$$

Subtract:

$$S_n - rS_n = u_1 - u_1 r^n$$

so

$$S_n(1 - r) = u_1(1 - r^n).$$

Infinite sum. If $|r| < 1$, then $r^n \rightarrow 0$ as $n \rightarrow \infty$, so $S_\infty = u_1/(1 - r)$.

2.4 Financial mathematics

Compound growth and decay are geometric in disguise.

$$A = P \left(1 + \frac{r}{m}\right)^{mt} \quad (\text{compound interest})$$

$$A = P(1 + r)^n \quad \text{if interest is compounded once per period}$$

$$A = P(1 - r)^n \quad \text{for repeated depreciation at rate } r$$

Exam note. Distinguish carefully between annual rate, monthly rate, and total number of compounding periods. Most errors here come from mismatched units, not algebra.

2.5 Sigma notation

Sigma notation is a compact way to express repeated addition:

$$\sum_{k=1}^n (3k + 2) = \sum_{k=1}^n 3k + \sum_{k=1}^n 2.$$

Useful expansions:

$$\sum_{k=1}^n k = \frac{n(n+1)}{2}, \quad \sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}, \quad \sum_{k=1}^n k^3 = \left(\frac{n(n+1)}{2}\right)^2$$

The last two are especially useful in calculus and statistics contexts.

2.6 Exponents and logarithms

$$a^m a^n = a^{m+n}, \quad \frac{a^m}{a^n} = a^{m-n}, \quad (a^m)^n = a^{mn}$$

$$a^{-n} = \frac{1}{a^n}, \quad a^{1/n} = \sqrt[n]{a}, \quad a^{m/n} = \sqrt[n]{a^m}$$

Logarithms are inverse exponentials:

$$\log_a x = y \iff a^y = x, \quad a > 0, a \neq 1, x > 0.$$

$$\log_a(xy) = \log_a x + \log_a y, \quad \log_a\left(\frac{x}{y}\right) = \log_a x - \log_a y, \quad \log_a(x^m) = m \log_a x$$

$$\log_a x = \frac{\log_b x}{\log_b a}$$

Why the laws work. If $x = a^u$ and $y = a^v$, then $xy = a^{u+v}$, so the product law follows directly from exponent laws.

For exponential equations, first isolate the power term. For logarithmic equations, first state the domain (argument > 0) before solving. If the base changes during manipulation, keep the base consistent until the end.

2.7 Proof

Direct proof. Start from the assumptions and transform logically to the conclusion.

Contradiction. Assume the negation of what you want to prove and show that this causes an impossibility.

Counterexample. One valid counterexample destroys a universal claim.

Induction. To prove $P(n)$ for all $n \geq n_0$:

- Base case: prove $P(n_0)$.
- Inductive step: assume $P(k)$, prove $P(k+1)$.
- Conclude by induction.

Induction template. If $1 + 2 + \dots + n = \frac{n(n+1)}{2}$ is required, assume true for k :

$$1 + 2 + \dots + k = \frac{k(k+1)}{2}.$$

Then

$$1 + 2 + \dots + k + (k+1) = \frac{k(k+1)}{2} + (k+1) = \frac{(k+1)(k+2)}{2}.$$

2.8 Binomial theorem

$$(a+b)^n = \sum_{r=0}^n \binom{n}{r} a^{n-r} b^r, \quad \binom{n}{r} = \frac{n!}{r!(n-r)!}$$

General term:

$$T_{r+1} = \binom{n}{r} a^{n-r} b^r$$

Core idea. Each term comes from choosing r copies of b and $n-r$ copies of a when expanding the product $(a+b)(a+b)\dots(a+b)$.

Extension to fractional/negative indices. For $|x| < 1$,

$$(1+x)^m = 1 + mx + \frac{m(m-1)}{2!}x^2 + \frac{m(m-1)(m-2)}{3!}x^3 + \dots$$

This is a power series expansion, not a finite polynomial.

2.9 Counting principles

$${}_nP_r = \frac{n!}{(n-r)!}, \quad {}_nC_r = \frac{n!}{r!(n-r)!}$$

When to use which. Permutations count ordered arrangements. Combinations count selections where order does not matter.

Common exam extensions. Circular arrangements give $(n-1)!$ for distinct objects in a circle, unless reflection symmetry also matters. With repeated objects, divide by factorials of repetitions.

2.10 Partial fractions

For rational functions whose denominator factors into linear factors, decompose:

$$\frac{P(x)}{(x-a)(x-b)} = \frac{A}{x-a} + \frac{B}{x-b}.$$

For repeated linear factors:

$$\frac{P(x)}{(x-a)^2} = \frac{A}{x-a} + \frac{B}{(x-a)^2}.$$

For irreducible quadratic factors:

$$\frac{P(x)}{x^2 + px + q} = \frac{Ax + B}{x^2 + px + q}.$$

Method. Multiply through by the denominator, compare coefficients, or substitute convenient values.

In integration questions, partial fractions is usually a setup skill. Most of the marks come from correct decomposition and sensible coefficient matching.

2.11 Complex numbers

$$i^2 = -1, \quad z = a + bi, \quad \bar{z} = a - bi, \quad |z| = \sqrt{a^2 + b^2}$$

The Argand diagram plots the real part on the horizontal axis and the imaginary part on the vertical axis.

Polar form.

$$z = r(\cos \theta + i \sin \theta) = r \operatorname{cis} \theta = re^{i\theta}$$

where $r = |z|$ and $\theta = \arg z$.

$$zw = r_1 r_2 \operatorname{cis}(\theta_1 + \theta_2), \quad \frac{z}{w} = \frac{r_1}{r_2} \operatorname{cis}(\theta_1 - \theta_2)$$

De Moivre.

$$(r \operatorname{cis} \theta)^n = r^n \operatorname{cis}(n\theta)$$

This gives roots of complex numbers:

$$z^{1/n} = r^{1/n} \operatorname{cis} \left(\frac{\theta + 2k\pi}{n} \right), \quad k = 0, 1, \dots, n-1.$$

For exact roots, never leave the answer as a decimal unless the question asks for approximation. Give all roots in the correct interval or in full set form.

2.12 Topic-summary exam strategy

When a Number and Algebra question looks long, it is usually testing one of five things: pattern recognition, algebraic manipulation, proof logic, accurate notation, or the ability to keep track of constraints like $|r| < 1$ and $x > 0$.

3 Functions

3.1 Definition and notation

A function maps each input to exactly one output. If $f : A \rightarrow B$, then $f(x)$ is the image of x under f .

Do not call something a function unless the mapping is single-valued. A vertical line test is a quick graph check, but the definition is the real test.

3.2 Domain and range

The domain is the set of allowed inputs. The range is the set of outputs actually attained.

Common restrictions:

$$\frac{1}{x} \Rightarrow x \neq 0, \quad \sqrt{x} \Rightarrow x \geq 0, \quad \log x \Rightarrow x > 0.$$

3.3 Straight lines

$$y = mx + c, \quad m = \frac{y_2 - y_1}{x_2 - x_1}$$

Parallel lines have equal gradients. Perpendicular lines satisfy $m_1 m_2 = -1$ when both gradients are defined.

3.4 Composite and inverse functions

$$(f \circ g)(x) = f(g(x)).$$

An inverse function undoes a function:

$$f^{-1}(f(x)) = x.$$

To find f^{-1} :

1. Set $y = f(x)$.
2. Swap x and y .
3. Solve for y .
4. State any domain restriction.

If a function is not one-to-one, restrict the domain before inverting. This is one of the easiest places to lose a clean mark.

3.5 Quadratic functions

$$f(x) = ax^2 + bx + c$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad \Delta = b^2 - 4ac$$

Three useful forms:

$$f(x) = ax^2 + bx + c, \quad f(x) = a(x - p)(x - q), \quad f(x) = a(x - h)^2 + k.$$

Why completing the square works.

$$ax^2 + bx + c = a \left(x^2 + \frac{b}{a}x \right) + c$$

Then

$$x^2 + \frac{b}{a}x = \left(x + \frac{b}{2a} \right)^2 - \left(\frac{b}{2a} \right)^2.$$

So the vertex form reveals the turning point directly.

3.6 Polynomial functions

Roots are the values of x that make $f(x) = 0$. A root gives a factor.

$$f(a) = 0 \iff (x - a) \text{ is a factor of } f(x)$$

$$f(x) = (x - a)q(x) + r, \quad r = f(a)$$

These are the factor theorem and remainder theorem. If a polynomial has roots α, β, γ , then the factorized form is

$$f(x) = k(x - \alpha)(x - \beta)(x - \gamma).$$

3.7 Rational functions

A rational function is a ratio of polynomials. Key features are asymptotes, intercepts, holes, and end behavior.

If

$$f(x) = \frac{ax + b}{cx + d},$$

then the vertical asymptote is $x = -d/c$ (if $c \neq 0$) and the horizontal asymptote is $y = a/c$.

For higher-degree rational functions, compare leading terms to infer asymptotes.

3.8 Exponential and logarithmic graphs

$$y = a^x, \quad y = e^x, \quad y = \log_a x, \quad y = \ln x.$$

Exponential graphs model growth/decay and logarithms are their inverses.

3.9 Transformations

From $y = f(x)$:

$$y = f(x) + b \text{ shifts up by } b,$$

$$y = f(x - a) \text{ shifts right by } a,$$

$$y = pf(x) \text{ stretches vertically by } |p|,$$

$$y = f(qx) \text{ compresses horizontally by factor } 1/|q|.$$

Reflections:

$$y = -f(x) \text{ across the } x\text{-axis}, \quad y = f(-x) \text{ across the } y\text{-axis}.$$

3.10 Odd and even functions

$$f(-x) = f(x) \Rightarrow \text{even}, \quad f(-x) = -f(x) \Rightarrow \text{odd}.$$

Even graphs are symmetric about the y -axis. Odd graphs are symmetric about the origin.

3.11 Inequalities

Solve graphically or algebraically. For a quadratic inequality, use a sign chart after factorisation or identify the region from the parabola.

A solid answer to an inequality question should state the solution set in interval notation or set notation, not just draw a sketch.

3.12 Special graphs

$$y = |f(x)|, \quad y = f(|x|), \quad y = \frac{1}{f(x)}, \quad y = [f(x)]^2.$$

Each one changes symmetry or sign structure. These are common Paper 1 and Paper 2 graph questions because they test interpretation, not just algebra.

4 Geometry and Trigonometry

4.1 Right-angled trigonometry

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}, \quad \cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}, \quad \tan \theta = \frac{\text{opposite}}{\text{adjacent}}$$

Pythagoras:

$$a^2 + b^2 = c^2.$$

4.2 Radians, arc length, and sector area

Radians are the natural angle unit because they make calculus and series cleaner.

$$\theta = \frac{s}{r}, \quad s = r\theta, \quad A = \frac{1}{2}r^2\theta$$

4.3 The unit circle and exact values

The unit circle gives all trigonometric values as coordinates and signs. Learn the special angles:

$$0, \frac{\pi}{6}, \frac{\pi}{4}, \frac{\pi}{3}, \frac{\pi}{2}$$

and use symmetry to extend to all quadrants.

4.4 Trigonometric identities

$$\begin{aligned} \sin^2 x + \cos^2 x &= 1 \\ 1 + \tan^2 x &= \sec^2 x, \quad 1 + \cot^2 x = \csc^2 x \\ \sin(2x) &= 2 \sin x \cos x \\ \cos(2x) &= \cos^2 x - \sin^2 x = 1 - 2 \sin^2 x = 2 \cos^2 x - 1 \\ \sin(\alpha \pm \beta) &= \sin \alpha \cos \beta \pm \cos \alpha \sin \beta \\ \cos(\alpha \pm \beta) &= \cos \alpha \cos \beta \mp \sin \alpha \sin \beta \end{aligned}$$

How to prove identities. Transform the more complicated side into the simpler side. Use known identities to replace everything in terms of sin and cos if needed.

4.5 Sine rule and cosine rule

For triangle ABC with sides a, b, c opposite angles A, B, C :

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

Area:

$$\Delta = \frac{1}{2}ab \sin C.$$

Ambiguous case. The sine rule can give two possible triangles because $\sin \theta = \sin(180^\circ - \theta)$.

4.6 Coordinate geometry in the plane and space

Distance in 2D:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}.$$

Midpoint:

$$M\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right).$$

4.7 Vectors

$$\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}, \quad |\mathbf{a}| = \sqrt{a_1^2 + a_2^2}$$

$$\mathbf{a} \cdot \mathbf{b} = a_1b_1 + a_2b_2 = |\mathbf{a}||\mathbf{b}|\cos\theta$$

For lines:

$$\mathbf{r} = \mathbf{a} + t\mathbf{d}.$$

Here $\mathbf{a}$ is a point vector and $\mathbf{d}$ is a direction vector.

Why the dot product matters. It converts geometry into algebra. Angles, perpendicularity, and projections all become manageable once written with dot products.

4.8 3D geometry

A plane can be written as

$$\mathbf{r} \cdot \mathbf{n} = c,$$

or in Cartesian form

$$ax + by + cz = d.$$

The vector $\mathbf{n} = (a, b, c)$ is normal to the plane.

Intersection idea. A line intersects a plane when substituting the parametric line equation into the plane gives a solvable linear equation in the parameter.

4.9 Topic-summary exam strategy

Geometry and Trigonometry problems often mix algebra, shape interpretation, and coordinate calculation. A strong solution usually states a diagram assumption, writes the relevant formula, substitutes carefully, and ends with a geometric conclusion such as angle, area, or position.

5 Statistics and Probability

5.1 Data, averages, and spread

Mean:

$$\bar{x} = \frac{\sum x}{n}$$

For frequency data:

$$\bar{x} = \frac{\sum fx}{\sum f}.$$

Variance and standard deviation:

$$\sigma^2 = \frac{\sum (x - \mu)^2}{n}, \quad s^2 = \frac{\sum (x - \bar{x})^2}{n - 1}.$$

Computational formula:

$$\sigma^2 = \frac{\sum x^2}{n} - \mu^2.$$

Do not mix population and sample formulas. The notation tells you which denominator is required.

5.2 Probability basics

$$P(A) = \frac{\text{favourable outcomes}}{\text{total outcomes}}$$

if outcomes are equally likely. For any event A :

$$0 \leq P(A) \leq 1, \quad P(A') = 1 - P(A).$$

5.3 Conditional probability and independence

$$P(A | B) = \frac{P(A \cap B)}{P(B)} \quad (P(B) > 0)$$

$$P(A \cap B) = P(A) P(B | A) = P(B) P(A | B)$$

$$A \text{ and } B \text{ independent} \iff P(A \cap B) = P(A)P(B)$$

5.4 Bayes' theorem

$$P(A_i | B) = \frac{P(B | A_i)P(A_i)}{\sum_j P(B | A_j)P(A_j)}.$$

This is the engine behind reverse-probability questions.

5.5 Counting in probability

Use permutations and combinations to count outcomes carefully before calculating probabilities. If a question says “at least” or “exactly”, translate the phrase into counting conditions before applying formulas.

5.6 Discrete random variables

A discrete random variable X assigns a number to each outcome.

$$E(X) = \sum x P(X = x)$$

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

Meaning. Expectation is the long-run average. Variance measures spread around the expectation.

5.7 Binomial distribution

A binomial setting has:

- a fixed number n of independent trials,
- only two outcomes per trial,
- constant success probability p .

$$X \sim B(n, p), \quad P(X = r) = \binom{n}{r} p^r (1-p)^{n-r}$$

$$E(X) = np, \quad \text{Var}(X) = np(1-p)$$

5.8 Normal distribution

$$X \sim N(\mu, \sigma^2).$$

Standardisation:

$$Z = \frac{X - \mu}{\sigma} \sim N(0, 1).$$

$$P(a < X < b) = P\left(\frac{a - \mu}{\sigma} < Z < \frac{b - \mu}{\sigma}\right)$$

Exam note. Always convert to Z before using a normal table or calculator function. For inverse normal questions, identify whether you are solving for x , μ , or σ .

5.9 Topic-summary exam strategy

Statistics and probability questions often reward a disciplined chain: define the variable, state the distribution, write the probability expression, and interpret the answer in context. The final line should answer the wording of the question, not just present a number.

6 Calculus

6.1 Limits and continuity

A limit describes what a function approaches. The formal definition is not usually required in AA HL, but the idea matters:

$$\lim_{x \rightarrow a} f(x) = L.$$

Continuity at $x = a$ means the function is defined there, the limit exists, and both match:

$$f(a) = \lim_{x \rightarrow a} f(x).$$

6.2 First principles

Derivative as a limit of average rate of change:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}.$$

This is the conceptual root of all differentiation formulas.

6.3 Differentiation rules

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

$$\frac{d}{dx}[af(x) + bg(x)] = af'(x) + bg'(x)$$

$$\frac{d}{dx}[f(x)g(x)] = f'(x)g(x) + f(x)g'(x)$$

$$\frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}$$

$$\frac{d}{dx}f(g(x)) = f'(g(x))g'(x)$$

Why the product rule works. Expand

$$[f(x) + \Delta f][g(x) + \Delta g] - f(x)g(x)$$

and ignore the small product $\Delta f \Delta g$ in the limit. The same logic yields the quotient and chain rules.

6.4 Standard derivatives

$$\frac{d}{dx}(\sin x) = \cos x, \quad \frac{d}{dx}(\cos x) = -\sin x, \quad \frac{d}{dx}(\tan x) = \sec^2 x$$

$$\frac{d}{dx}(e^x) = e^x, \quad \frac{d}{dx}(a^x) = a^x \ln a, \quad \frac{d}{dx}(\ln x) = \frac{1}{x}$$

Inverse trig derivatives:

$$\frac{d}{dx}(\arcsin x) = \frac{1}{\sqrt{1-x^2}}, \quad \frac{d}{dx}(\arccos x) = -\frac{1}{\sqrt{1-x^2}}, \quad \frac{d}{dx}(\arctan x) = \frac{1}{1+x^2}.$$

6.5 Implicit differentiation

Differentiate both sides with respect to x , treating y as a function of x . For example, if $x^2 + y^2 = 25$, then

$$2x + 2y \frac{dy}{dx} = 0$$

so

$$\frac{dy}{dx} = -\frac{x}{y}.$$

6.6 Second derivative and curve sketching

$$f''(x) = \frac{d^2y}{dx^2}$$

tests concavity:

- $f''(x) > 0$ means concave up,
- $f''(x) < 0$ means concave down.

Stationary points satisfy

$$f'(x) = 0.$$

Classify them using the sign of $f''(x)$ or by a sign change in $f'(x)$.

6.7 Tangents and normals

At $x = a$:

$$\text{tangent: } y - f(a) = f'(a)(x - a)$$

$$\text{normal gradient} = -\frac{1}{f'(a)} \quad \text{if } f'(a) \neq 0.$$

6.8 Applications of differentiation

Rates of change. If $s(t)$ is displacement, then

$$v(t) = \frac{ds}{dt}, \quad a(t) = \frac{dv}{dt} = \frac{d^2s}{dt^2}.$$

Optimisation. Define the quantity to be maximised/minimised, write it as a function of one variable, differentiate, solve $f'(x) = 0$, test, and then interpret the answer.

Related rates. Differentiate an equation linking variables with respect to time, then substitute numerical values after differentiating.

6.9 Integration

Integration reverses differentiation:

$$\int f'(x) \, dx = f(x) + C.$$

$$\int x^n \, dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1)$$

$$\int \frac{1}{x} \, dx = \ln |x| + C$$

$$\int e^x \, dx = e^x + C$$

$$\int \sin x \, dx = -\cos x + C, \quad \int \cos x \, dx = \sin x + C$$

Definite integral:

$$\int_a^b f(x) \, dx = F(b) - F(a),$$

where $F'(x) = f(x)$.

6.10 Area and volume

Area under a curve is usually obtained from a definite integral; area between curves is

$$\int_a^b [f(x) - g(x)] \, dx$$

when $f(x) \geq g(x)$ on $[a, b]$.

Volume of revolution about the x -axis:

$$V = \pi \int_a^b [f(x)]^2 \, dx.$$

6.11 Substitution and integration by parts

Substitution:

$$\int f(g(x))g'(x) \, dx = \int f(u) \, du.$$

Integration by parts:

$$\int u \, dv = uv - \int v \, du.$$

Choosing u . Usually choose the function that simplifies when differentiated. A common rule of thumb is LIATE: logarithmic, inverse trig, algebraic, trig, exponential.

6.12 Differential equations

A differential equation relates a function to its derivatives.

Separable form.

$$\frac{dy}{dx} = g(x)h(y)$$

rearranges to

$$\frac{1}{h(y)} \, dy = g(x) \, dx$$

and then integrate both sides.

Growth/decay model.

$$\frac{dN}{dt} = kN \quad \Rightarrow \quad N = N_0 e^{kt}.$$

6.13 Exam strategy for calculus

The fastest route to marks is usually:

1. define the function cleanly,
2. differentiate or integrate with correct notation,
3. show the algebra line-by-line,
4. interpret the result in context,
5. check whether the answer should be exact or approximate.

7 Derivation gallery

7.1 Quadratic formula from completing the square

Start with

$$ax^2 + bx + c = 0, \quad a \neq 0.$$

Divide by a :

$$x^2 + \frac{b}{a}x + \frac{c}{a} = 0.$$

Move the constant term:

$$x^2 + \frac{b}{a}x = -\frac{c}{a}.$$

Complete the square:

$$x^2 + \frac{b}{a}x + \left(\frac{b}{2a}\right)^2 = \left(x + \frac{b}{2a}\right)^2.$$

Add the same term to the right side:

$$\left(x + \frac{b}{2a}\right)^2 = \frac{b^2 - 4ac}{4a^2}.$$

Taking square roots gives

$$x + \frac{b}{2a} = \pm \frac{\sqrt{b^2 - 4ac}}{2a},$$

hence

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

7.2 Arithmetic series from pairing

Let

$$S_n = u_1 + u_2 + \cdots + u_n.$$

For an arithmetic sequence, $u_n = u_1 + (n-1)d$. Writing the sum in reverse gives

$$S_n = u_n + u_{n-1} + \cdots + u_1.$$

Add the two:

$$2S_n = (u_1 + u_n) + (u_2 + u_{n-1}) + \cdots.$$

Every pair sums to $u_1 + u_n$, and there are n pairs, so

$$2S_n = n(u_1 + u_n).$$

This is the cleanest derivation to remember because it explains why the formula is symmetric in first and last term.

7.3 Geometric series from subtraction

Let

$$S_n = u_1 + u_1r + u_1r^2 + \cdots + u_1r^{n-1}.$$

Then

$$rS_n = u_1r + u_1r^2 + \cdots + u_1r^n.$$

Subtracting:

$$S_n - rS_n = u_1 - u_1r^n.$$

So

$$S_n(1 - r) = u_1(1 - r^n).$$

The infinite series formula follows only when $|r| < 1$, because then $r^n \rightarrow 0$.

7.4 Binomial theorem by induction

The statement

$$(1+x)^n = \sum_{r=0}^n \binom{n}{r} x^r$$

can be proved by induction.

Base case: for $n = 1$, the statement is clear.

Inductive step: assume true for $n = k$:

$$(1+x)^k = \sum_{r=0}^k \binom{k}{r} x^r.$$

Then

$$(1+x)^{k+1} = (1+x) \sum_{r=0}^k \binom{k}{r} x^r.$$

Distribute:

$$(1+x)^{k+1} = \sum_{r=0}^k \binom{k}{r} x^r + \sum_{r=0}^k \binom{k}{r} x^{r+1}.$$

Shift indices in the second sum and combine coefficients using Pascal's identity

$$\binom{k}{r} + \binom{k}{r-1} = \binom{k+1}{r}.$$

That gives the next row of the theorem.

7.5 Logarithm laws from exponent laws

Let $x = a^u$ and $y = a^v$. Then $xy = a^{u+v}$, so

$$\log_a(xy) = u + v = \log_a x + \log_a y.$$

Similarly,

$$\log_a\left(\frac{x}{y}\right) = u - v = \log_a x - \log_a y.$$

For powers,

$$x^m = (a^u)^m = a^{um},$$

so

$$\log_a(x^m) = um = m \log_a x.$$

The laws are therefore not separate facts; they are consequences of exponents and the inverse relationship between exponentials and logarithms.

7.6 Quadratic roots and the discriminant

The discriminant

$$\Delta = b^2 - 4ac$$

tells you how many real roots a quadratic has:

- $\Delta > 0$: two distinct real roots;
- $\Delta = 0$: one repeated real root;
- $\Delta < 0$: no real roots.

This comes directly from the square-root term in the quadratic formula. If the square root is not real, the roots are not real.

7.7 Sine rule from area

Area of a triangle can be written in three ways:

$$\Delta = \frac{1}{2}bc \sin A = \frac{1}{2}ca \sin B = \frac{1}{2}ab \sin C.$$

Equating the first two:

$$\frac{1}{2}bc \sin A = \frac{1}{2}ca \sin B.$$

Cancel $\frac{1}{2}c$:

$$b \sin A = a \sin B.$$

Rearranging gives

$$\frac{a}{\sin A} = \frac{b}{\sin B}.$$

Repeating this with the other pairs yields the sine rule.

7.8 Cosine rule from coordinates

Place triangle ABC in the plane with

$$A = (0, 0), \quad B = (c, 0), \quad C = (b \cos A, b \sin A).$$

Then

$$BC^2 = (c - b \cos A)^2 + (0 - b \sin A)^2.$$

Expanding:

$$BC^2 = c^2 - 2bc \cos A + b^2 \cos^2 A + b^2 \sin^2 A.$$

Use $\cos^2 A + \sin^2 A = 1$:

$$BC^2 = a^2 = b^2 + c^2 - 2bc \cos A.$$

This is the cosine rule.

7.9 Dot product from coordinates

For vectors $\mathbf{u} = (u_1, u_2)$ and $\mathbf{v} = (v_1, v_2)$, the dot product is

$$\mathbf{u} \cdot \mathbf{v} = u_1 v_1 + u_2 v_2.$$

If the angle between them is θ , then

$$\mathbf{u} \cdot \mathbf{v} = |\mathbf{u}| |\mathbf{v}| \cos \theta.$$

The coordinate formula and the geometric formula are consistent because the projections of one vector onto the other multiply one component by the cosine of the included angle.

7.10 Product, quotient, and chain rule proof idea

These rules all come from the same limit logic. Let increments be Δf and Δg .

$$(fg)(x+h) - f(x)g(x) = f(x)\Delta g + g(x)\Delta f + \Delta f \Delta g.$$

After dividing by h and taking $h \rightarrow 0$, the product $\Delta f \Delta g/h$ vanishes under smoothness assumptions, leaving the product rule. The quotient rule is a similar rearrangement. The chain rule arises because the inside increment gets multiplied by the outside derivative.

7.11 Integration by parts from the product rule

Start with the product rule:

$$\frac{d}{dx}(uv) = u \frac{dv}{dx} + v \frac{du}{dx}.$$

Integrate both sides:

$$uv = \int u \frac{dv}{dx} dx + \int v \frac{du}{dx} dx.$$

Rearrange:

$$\int u dv = uv - \int v du.$$

This is the formula you use whenever direct integration is awkward but differentiating one factor makes the integrand simpler.

7.12 Standardisation of the normal distribution

If $X \sim N(\mu, \sigma^2)$ and

$$Z = \frac{X - \mu}{\sigma},$$

then Z has mean 0 and standard deviation 1. This transformation rescales and recentres the entire distribution. All normal probability questions are therefore reduced to the standard normal curve.

7.13 Bayes theorem from conditional probability

From conditional probability:

$$P(A_i | B) = \frac{P(A_i \cap B)}{P(B)}.$$

But

$$P(A_i \cap B) = P(B | A_i)P(A_i).$$

So

$$P(A_i | B) = \frac{P(B | A_i)P(A_i)}{P(B)}.$$

Then expand the denominator using the law of total probability:

$$P(B) = \sum_j P(B | A_j)P(A_j).$$

This gives Bayes' theorem.

8 Common exam pitfalls

8.1 Algebra

A lot of marks disappear through one of five mistakes: sign errors, dropped brackets, incorrect factorisation, illegal cancellation, or stopping one line too early. If the paper asks for “hence”, your previous line must genuinely support the next one.

8.2 Functions and graphs

State the domain before discussing inverse functions or logarithms. When sketching a graph, label asymptotes, intercepts, turning points, and end behavior. A sketch without features is often incomplete even if it looks visually correct.

8.3 Trigonometry

Always pay attention to radians versus degrees. In derivations, use exact values until the very end. For inverse trig, remember the restricted output range.

8.4 Calculus

Do not differentiate and forget the constant of differentiation if you are reversing the process. For definite integrals, the numerical answer must be interpreted with the correct sign. For optimisation, verify that the stationary point is actually a maximum or minimum.

8.5 Probability and statistics

Define random variables explicitly. Keep n , μ , σ , $\bar{x}$, and s distinct. In conditional probability, do not confuse $P(A | B)$ with $P(B | A)$.

9 Additional enrichment: matrices and determinants

9.1 Matrices

A matrix is a rectangular array of numbers. For AA HL, matrices are not a central syllabus topic, but they are useful enrichment and can appear in wider mathematical study.

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

For 2×2 matrices:

$$\det(A) = ad - bc, \quad A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

when $ad - bc \neq 0$.

9.2 Solving linear systems

A system of equations can be written as $AX = B$. If A is invertible, then

$$X = A^{-1}B.$$

This is the matrix version of elimination. It is useful conceptually because it shows why some systems have a unique solution and some do not.

9.3 Determinants and geometry

A determinant gives an area scaling factor in 2D and a volume scaling factor in 3D. If the determinant is zero, the transformation collapses dimension and the matrix is singular.

10 Final command-term guide

State	Give a result with minimal explanation.
Show that	Present the algebraic steps that prove the claim.
Hence	Use a previous result directly.
Hence or otherwise	Use any valid method, but try to build on earlier parts.
Determine	Find the value from the given information and show reasoning.
Prove	Provide a logically complete argument.
Explain	Give the reasoning, not just the result.

11 Last word for the exam

The best AA HL answers are not the most complicated ones. They are the ones where every line has a purpose. If you can see the structure of the argument before you start writing, the examiner usually can too.

12 IB exam technique

12.1 Paper 1

No calculator means algebra must be clean. Learn to manipulate surds, exponents, logs, and exact trig values fluently. If the answer looks ugly, the question may still be expecting an exact form.

12.2 Paper 2

Use technology to confirm graphs, roots, intersections, table values, regression, and distribution outputs, but do not let the calculator replace your reasoning. The written method still matters.

12.3 Paper 3

Paper 3 often rewards structured thinking over speed. Write clearly enough that the path from the question to the conclusion is visible. If there are multiple parts, use each previous result deliberately; that is often what the paper is designed to test.

12.4 Internal Assessment

Choose a mathematical exploration with depth, not just a topic with interesting context. The IA needs a question, a model, mathematics that goes beyond routine calculation, reflection on limitations, and a clean conclusion.

The difference between a 5 and a 7 is often not advanced content. It is tighter structure, fewer algebra slips, better labels, and more deliberate interpretation.

13 Ultra-condensed final formula sheet

$$u_n = u_1 + (n - 1)d$$

$$S_n = \frac{n}{2}(2u_1 + (n - 1)d)$$

$$u_n = u_1 r^{n-1}$$

$$S_n = \frac{u_1(1 - r^n)}{1 - r}$$

$$S_\infty = \frac{u_1}{1 - r}$$

$$\binom{n}{r} = \frac{n!}{r!(n - r)!}$$

$$(a + b)^n = \sum_{r=0}^n \binom{n}{r} a^{n-r} b^r$$

$$A = P(1 + r)^n$$

$$\log_a(xy) = \log_a x + \log_a y$$

$$\sin^2 x + \cos^2 x = 1$$

$$\sin(2x) = 2 \sin x \cos x$$

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}| \cos \theta$$

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

$$X \sim B(n, p), \quad P(X = r) = \binom{n}{r} p^r (1 - p)^{n-r}$$

$$Z = \frac{X - \mu}{\sigma}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x + h) - f(x)}{h}$$

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

$$V = \pi \int_a^b [f(x)]^2 dx$$